

JEE MAIN REVISION MASTER: HEAT AND THERMODYNAMICS SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Heat And Thermodynamics - Core Application	28 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Heat And Thermodynamics - Core Application (Very High Yield — 28 PYQs)

Core Principle: Key problem pattern in Heat And Thermodynamics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 4..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : If the average kinetic en..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K . Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K . The final pressure in the vessels is ____ kPa .

- (A) 100 **[Correct]**
- (B) 120
- (C) 90
- (D) 105

Correct Answer: A

Solution Analysis: Using Ideal Gas Law: $PV = nRT$ Where P is pressure, V is volume, n is number of moles, R is the gas constant, and T is absolute temperature. Since the system is closed, the total number of moles (n_{total}) remains constant before and after heating. Initial State: Both vessels (each of volume V) are at $P_{(0)}=90\text{kPa}$ and $T_{(0)}=400\text{K}$.
 $n_{\text{total}} = \frac{P_{(0)}V}{RT_{(0)}} + \frac{P_{(0)}V}{RT_{(0)}} = \frac{2P_{(0)}V}{RT_{(0)}}$... (i) Final State: Pressure becomes $P_{(f)}$ in both vessels. Vessel 1 stays at $T_{(1)}=400\text{K}$, and vessel 2 is at $T_{(2)}=500\text{K}$. $n_{\text{total}} = \frac{P_{(f)}V}{RT_{(1)}} + \frac{P_{(f)}V}{RT_{(2)}}$